DEVOPS FUNDAMENTALS
MINI PROJECT REPORT

Cloud-Based Linux Server Deployment with GitHub Version Control and Automation

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Batch Code: 530

Platform: AWS EC2, Ubuntu Linux, Apache2, Git/GitHub

Project Date: 18 October 2026

1. Project Overview

This project demonstrates foundational DevOps practices by deploying a website on an AWS EC2 Ubuntu server, managing files with Git and GitHub, automating administration tasks with Bash scripts, creating timestamped website backups, and reviewing system and authentication logs.

2. Objectives

• Provision and access a Linux server in AWS.
• Install and manage the Apache2 web service.
• Publish a website using the server's public address.
• Use Git/GitHub for version control and remote storage.
• Automate health checks, service monitoring, and website backups.
• Generate a log report containing authentication activity, system errors, and warnings.

3. AWS Architecture

Developer / Git client → GitHub repository → AWS EC2 Ubuntu instance → Apache2 → Website

The EC2 security group controls inbound network access. SSH is used for remote administration, while HTTP traffic reaches Apache on port 80. For safer operation, restrict SSH access to trusted source IP addresses rather than opening it to everyone.

4. Implementation Steps

1. Launch an Ubuntu EC2 instance and connect over SSH.
2. Update package metadata and install Apache2 and required command-line tools.
3. Place the website file at the Apache document root.
4. Create Bash scripts for health checks, Apache monitoring, and website backup.
5. Generate a log report from the server's available logs.
6. Track project files with Git, commit changes, and push them to GitHub.
7. Capture screenshots as evidence of setup and script output.

5. Automation Scripts

server-health.sh: reports information such as CPU, memory, disk usage, running processes, and uptime.
apache-monitor.sh: checks Apache2 service status and is intended to restart it if stopped.
website-backup.sh: creates a timestamped compressed website backup in the Backup directory.
generate-log-report.sh: collects available log information into log-report.txt, including failed login attempts, SSH activity, system errors, and warnings.

6. Git and GitHub

Git was used to track project changes, create commits, and push the main branch to the remote GitHub repository. The latest reported push completed successfully.
Repository: https://github.com/khushiiii05/DevOps-Project

7. Challenges Faced

• Learning Linux commands, file permissions, and service management.
• Editing and organizing project files on the server.
• Configuring Git authentication and pushing changes to GitHub.
• Understanding authentication and system logs.
• Creating repeatable backups with timestamps.

8. Learning Outcomes

The project provided practical experience with Linux administration, AWS EC2, Apache2, Bash scripting, log inspection, compressed backups, and Git/GitHub version control. It also reinforced the importance of checking service status and securing remote access.

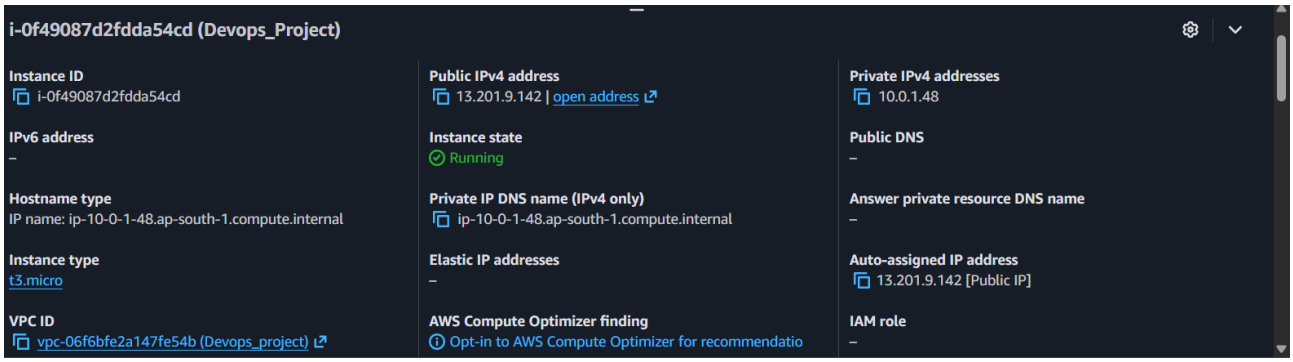
9. Conclusion

This mini project combines core introductory DevOps tasks in a small cloud-hosted environment. Future improvements could include scheduled backups, alerting, stricter security group rules, and automated deployment.

Appendix A. Project Screenshots

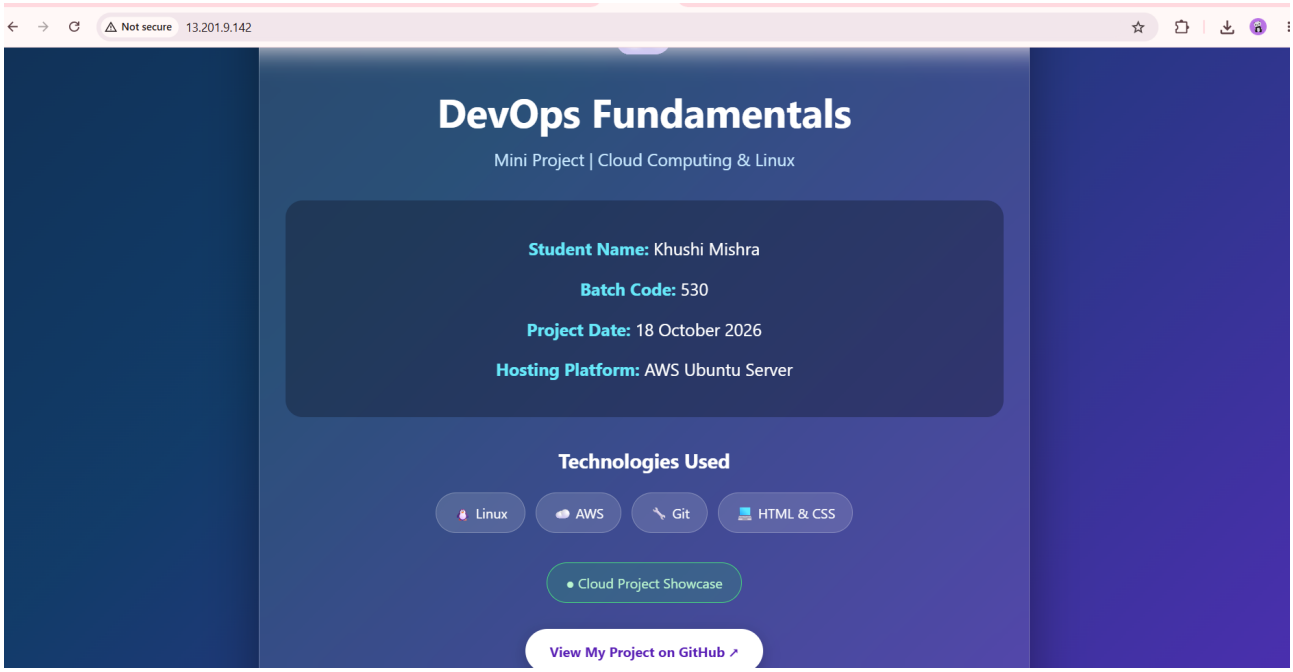
Screenshots supplied by the student are included below as evidence of setup and testing.

AWS EC2 and networking setup



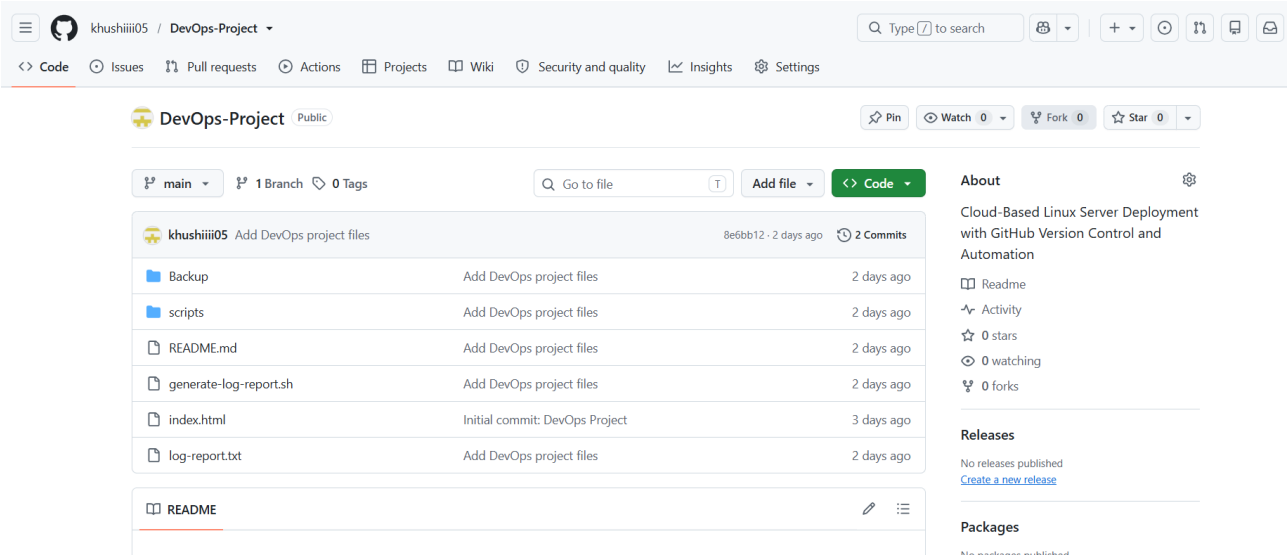
aws-setup.png.png

Website hosted on the EC2 server



website.png.png

GitHub repository



Github.png.png

Project scripts

```
ubuntu@ip-10-0-1-48:~$ cd ~/DevOps-Project
ls -l scripts/
total 12
-rwxrwxr-x 1 ubuntu ubuntu 646 Oct 19 10:06 apache-monitor.sh
-rwxrwxr-x 1 ubuntu ubuntu 467 Oct 19 09:59 server-health.sh
-rwxrwxr-x 1 ubuntu ubuntu 790 Oct 19 10:18 website-backup.sh
ubuntu@ip-10-0-1-48:~/DevOps-Project$
```

scripts.png.png

Server health check

```
ubuntu@ip-10-0-1-48:~$ cd ~/DevOps-Project
bash scripts/server-health.sh
================================
      SERVER HEALTH CHECK
================================

1. CPU Usage
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 95.5 id,  4.5 wa,  0.0 hi,  0.0 si,  0.0 st

2. Memory Usage
               total        used        free      shared  buff/cache   available
Mem:           908Mi       313Mi       439Mi       3.0Mi       264Mi       595Mi
Swap:             0B          0B          0B

3. Disk Usage
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.7G  4.0G  40% /
tmpfs            455M     0  455M   0% /dev/shm
tmpfs            182M  896K  181M   1% /run
efivarfs         128K  3.3K  120K   3% /sys/firmware/efi/efivars
tmpfs            455M     0  455M   0% /tmp
none             1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none             1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  989M  163M  759M  18% /boot
/dev/nvme0n1p15  105M  6.3M   99M   7% /boot/efi
none             1.0M     0  1.0M   0% /run/credentials/systemd-networkd.service
none             1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
none             1.0M     0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs             91M  8.0K   91M   1% /run/user/1000

4. Logged-in Users
```

server-health.png

Apache service monitor

```
ubuntu@ip-10-0-1-48:~/DevOps-Project$ bash scripts/apache-monitor.sh
=================================
     APACHE SERVICE MONITOR
=================================
SUCCESS: Apache is running.
```

apache-monitor.png

Website backup

```
ubuntu@ip-10-0-1-48:~/DevOps-Project$ cd ~/DevOps-Project
bash scripts/website-backup.sh
================================
      WEBSITE BACKUP
================================
SUCCESS: Website backup created!
Backup file: /home/ubuntu/DevOps-Project/Backup/website-backup-2026-10-22_09-10-13.tar.gz
4.0K    /home/ubuntu/DevOps-Project/Backup/website-backup-2026-10-22_09-10-13.tar.gz
ubuntu@ip-10-0-1-48:~/DevOps-Project$ ls -lh Backup/
total 8.0K
-rw-rw-r-- 1 ubuntu ubuntu 1.6K Oct 19 10:18 website-backup-2026-10-19_10-18-40.tar.gz
-rw-rw-r-- 1 ubuntu ubuntu 2.0K Oct 22 09:10 website-backup-2026-10-22_09-10-13.tar.gz
```

backup.png

Log report

```
ubuntu@ip-10-0-1-48:~/DevOps-Project$ cat log-report.txt
=======================================
      LINUX LOG MONITORING REPORT
Generated: Thu Oct 22 09:19:26 UTC 2026
Hostname: ip-10-0-1-48
=======================================

1. FAILED LOGIN ATTEMPTS
---------------------------------------
2026-10-19T11:03:10.553240+00:00 ip-10-0-1-48 sshd-session[2859]: Invalid user admin from 106.104.115.199 port 34306
2026-10-19T11:03:10.800456+00:00 ip-10-0-1-48 sshd-session[2859]: Connection closed by invalid user admin 106.104.115.199 port 34306 [preauth]
2026-10-19T11:03:13.547882+00:00 ip-10-0-1-48 sshd-session[2865]: Invalid user user from 106.104.115.199 port 34364
2026-10-19T11:03:13.783901+00:00 ip-10-0-1-48 sshd-session[2865]: Connection closed by invalid user user 106.104.115.199 port 34364 [preauth]
2026-10-19T11:03:14.638297+00:00 ip-10-0-1-48 sshd-session[2867]: Invalid user admin from 106.104.115.199 port 34384
2026-10-19T11:03:14.904868+00:00 ip-10-0-1-48 sshd-session[2867]: Connection closed by invalid user admin 106.104.115.199 port 34384 [preauth]
2026-10-19T11:03:16.772690+00:00 ip-10-0-1-48 sshd-session[2871]: Invalid user admin from 106.104.115.199 port 34420
2026-10-19T11:03:17.030446+00:00 ip-10-0-1-48 sshd-session[2871]: Connection closed by invalid user admin 106.104.115.199 port 34420 [preauth]
2026-10-19T11:03:18.910525+00:00 ip-10-0-1-48 sshd-session[2875]: Invalid user admin from 106.104.115.199 port 34460
2026-10-19T11:03:19.166475+00:00 ip-10-0-1-48 sshd-session[2875]: Connection closed by invalid user admin 106.104.115.199 port 34460 [preauth]
2026-10-19T11:03:29.261873+00:00 ip-10-0-1-48 sshd-session[2895]: Invalid user admin from 106.104.115.199 port 34782
2026-10-19T11:03:29.503673+00:00 ip-10-0-1-48 sshd-session[2895]: Connection closed by invalid user admin 106.104.115.199 port 34782 [preauth]
2026-10-19T11:03:30.299605+00:00 ip-10-0-1-48 sshd-session[2897]: Invalid user admin from 106.104.115.199 port 34808
2026-10-19T11:03:30.547518+00:00 ip-10-0-1-48 sshd-session[2897]: Connection closed by invalid user admin 106.104.115.199 port 34808 [preauth]
```

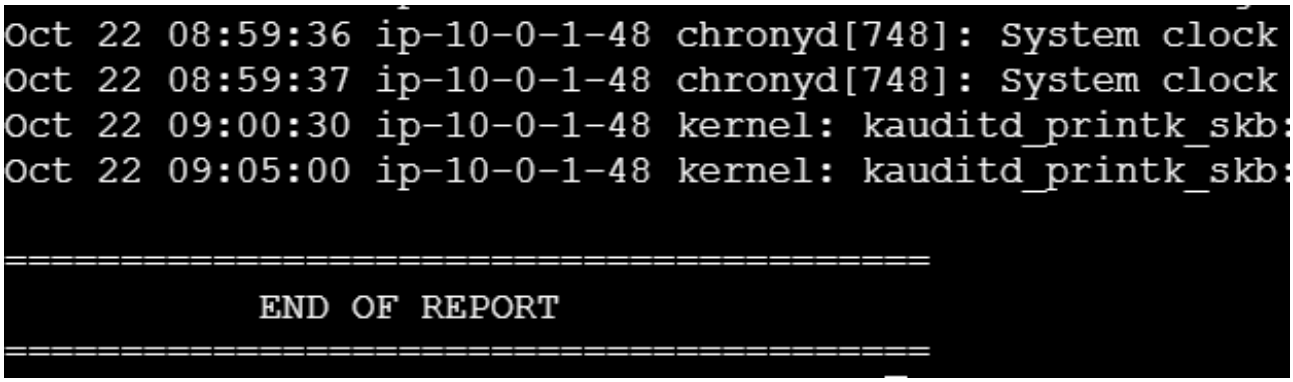
log-report.png

Additional log report evidence

```
2. SSH LOGIN ACTIVITY
----------------------------------------
2026-10-19T10:31:10.941775+00:00 ip-10-0-1-48 sudo: ubuntu : TTY=/dev/pts/0 ; PWD=/home/ubuntu/DevOps-Project ; USER=root ; COMMAND=/usr/bin/grep -Ei sshd.*(accepted|ses
sion opened|session closed) /var/log/auth.log
2026-10-19T11:32:26.914447+00:00 ip-10-0-1-48 sshd-session[1107]: pam_unix(sshd:session): session closed for user ubuntu
2026-10-19T12:57:30.503868+00:00 ip-10-0-1-48 sshd-session[3183]: Accepted publickey for ubuntu from 13.233.177.3 port 42392 ssh2: ED25519 SHA256:GMxYhBKQcZ8N3GJStFSkKoc
fpc4PUPBJ2Nxqtty6FtU
2026-10-19T12:57:30.505690+00:00 ip-10-0-1-48 sshd-session[3183]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
2026-10-19T13:16:25.309543+00:00 ip-10-0-1-48 sshd-session[3183]: pam_unix(sshd:session): session closed for user ubuntu
2026-10-21T09:13:32.494914+00:00 ip-10-0-1-48 sshd-session[1065]: Accepted publickey for ubuntu from 13.233.177.4 port 38111 ssh2: ED25519 SHA256:x+J5Uvz74zmPu3r7i5tmNwb
Bo5EobH4JfLuuFWqSLRQ
2026-10-21T09:13:32.498391+00:00 ip-10-0-1-48 sshd-session[1065]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
2026-10-21T10:16:08.581828+00:00 ip-10-0-1-48 sshd-session[1065]: pam_unix(sshd:session): session closed for user ubuntu
2026-10-21T10:18:35.909970+00:00 ip-10-0-1-48 sshd-session[2460]: Accepted publickey for ubuntu from 13.233.177.5 port 54238 ssh2: ED25519 SHA256:shhWYPdDILvhZIA5DeWoLFx
Pwt/0PD8mqySL4IM+heo
2026-10-21T10:18:35.911746+00:00 ip-10-0-1-48 sshd-session[2460]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
```

log-report2.png

Additional log report evidence



log-report3.png